

Outage Investigation Report —

OIR-2025-009

Field	Value
Document ID	OIR-2025-009
Report Date	2025-12-18
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Reviewing Manager	Priya Ramachandran, Plant Manager, NW-PowerPool
Region	NW-PowerPool
Site	NW-PowerPool-WIND-024
Asset	WIN-10011 (Siemens Gamesa, wind_turbine)
Outage ID	OUT-2025-0319
Classification	Forced outage — efficiency_decline
Distribution	NW-PowerPool Operations, Reliability Engineering, OEM Liaison, Compliance

1. Outage Summary

WIN-10011 declared a forced outage on 2025-12-02 at 05:58 PST after the unit's performance fell below the MS-2025-03 efficiency-decline threshold (8% below model-predicted power curve over a 30-day rolling window). The asset returned to service on 2025-12-06 at 20:58 PST, duration **111 hours**. Lost generation: **2,706.6 MWh**.

WIN-10011 has now experienced **12 forced outages and 14,411 MWh of lost generation in FY2025**, making it the second-worst-performing turbine in the Cascadia fleet (only WIN-10130 at CAISO-WIND-026 is worse — see OIR-2025-010 and RCA-2025-003).

2. Timeline of Events

Timestamp (PST)	Event
2025-11-02	30-day rolling efficiency analytic indicates WIN-10011 trending below model curve at high cut-in wind speeds. Variance -4.2%.
2025-11-15	Variance -5.8%. Operations flags for engineering review.
2025-12-01 21:30	Variance crosses -8.0% threshold. Continuous operation permitted pending diagnostic review.
2025-12-02 05:58	Variance touches -9.4%; SCADA initiates forced derate then planned shutdown sequence. Forced outage declared.
2025-12-02 08:00	CPS unavailable within 4 hours due to concurrent NW-PowerPool-WIND-003 transformer work (TRA-10561, see OIR-2025-004). EPA-2024-11 invoked.
2025-12-02 12:50	Apex onsite. Initial inspection: pitch-actuator hydraulic pressure on blade B reading 188 bar (spec 220-240). Blade-B pitch angle drifting 1.4° behind setpoint at high yaw rates.
2025-12-03 — 2025-12-05	Hydraulic accumulator replacement, pitch-system gasket rebuild, pitch encoder recalibration, blade-B re-pitching. Siemens Gamesa OEM provides remote advisory.
2025-12-06	Re-commissioning and power-curve validation. Post-repair variance: -0.7%.
2025-12-06 20:58	Returned to service.

3. Telemetry Observations

- Blade-B pitch-actuator hydraulic pressure: 188 bar (spec 220-240). Bench-tested accumulator showed pre-charge 41 bar (spec 80 bar). Slow leak over ~4 months.
- Pitch-angle tracking error blade B: 1.4° lag at high yaw rates (spec < 0.3°).
- Vibration RMS: 5.1 mm/s (within MS-2025-03 6.5 limit).
- Gearbox sump oil temperature: normal range; this is **not** an overheating event.
- Iron particulate in oil: 9 ppm — within limits (limit 12).
- CSM-7 pitch-system quarterly inspection (WO-2025-09204) was closed as "complete" on 2025-09-23 by Pinnacle Energy Maintenance. Hydraulic accumulator pre-charge check is listed in the procedure but not documented in close-out. See CAR-2025-004.

4. Restoration Activities

Vendor	Scope	Invoice (USD)
Apex Emergency Grid Solutions	Onsite diagnostics, project management	\$112,000
Apex / Siemens Gamesa subcontract (1.8x)	Hydraulic accumulator replacement, pitch-system gasket rebuild, encoder recalibration	\$209,000
Total		\$321,000

CPS scheduled equivalent estimate (post-event benchmark): \$87,400.

5. Preliminary Root Cause

Direct cause: slow nitrogen leak in blade-B pitch-actuator hydraulic accumulator over an extended period, leading to under-pressurized pitch response and aerodynamic mis-trim driving the efficiency decline.

Contributing causes:

1. CSM-7 pitch-system quarterly inspection (WO-2025-09204) close-out lacked documented pre-charge check. This is the third Pinnacle close-out documentation issue at NW-PowerPool in FY2025 (see also OIR-2025-001 and OIR-2025-003). Refer to CAR-2025-004.
2. The efficiency-decline trend was visible in the 30-day rolling analytic from 2025-11-02 onward. Engineering review was paused while the team responded to the Fall Overheating Sequence (OIR-2025-002, OIR-2025-003, OIR-2025-008). FMP-2 fatigue concerns implicated. See CAR-2025-001.
3. WIN-10011 is one of the four NW-PowerPool-WIND-024 assets covered by MOC-2025-022 (pending OEM threshold override). The Siemens Gamesa OEM bulletin SG-PB-2025-02 recommends pitch accumulator pre-charge verification every 6 months — Cascadia CSM-7 is on a 3-month cycle, which is correct, but the close-out documentation has been incomplete.

6. Recommendations

1. Execute pre-charge audit of all NW-PowerPool pitch accumulators within 30 days. Track under CAR-2025-004.
2. Re-validate Pinnacle close-out documentation at NW-PowerPool-WIND-024 and -003. Refer to CAR-2025-004.

3. Add efficiency-decline early-warning to the on-shift escalation matrix. Currently advisory; should be action-required at -5%.
4. Add WIN-10011 to the FY26 capital planning watchlist alongside WIN-10131 and WIN-10130. Refer to RCA-2025-003.
5. Cross-reference OIR-2025-002 (WIN-10131 August event), OIR-2025-008 (SUB-10629 November event), RCA-2025-002 (Fall Overheating Sequence), RCA-2025-003 (WIN-10130 chronic).

Signatures

Role	Name	Date
Author	Elena Vasquez, Reliability Engineer II	2025-12-18
Plant Manager	Priya Ramachandran	2025-12-19
OEM Liaison	J. Bertelsen	2025-12-19

Distribution: NW-PowerPool Operations · Reliability Engineering · OEM Liaison · Compliance · Vendor Management · CFO Office

References: OIR-2025-002 · OIR-2025-003 · OIR-2025-004 · OIR-2025-008 · CAR-2025-001 · CAR-2025-004 · MOC-2025-022 · RCA-2025-002 · RCA-2025-003 · RCA-2025-005 · Siemens Gamesa OEM Manual Rev. 4.2 · Bulletin SG-PB-2025-02 · Standard MS-2025-03 · CSM-7